

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data integration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks:20

Annexure A

CLASS TEST

<i>Criteria</i>	Understanding of Task (2)	Procedure & Execution (2.5)	Observation & Result (2)	Viva Voce / Conceptual Response (2)	Time Management & Presentation (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	20%	20%	15%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						
<i>Q6</i>						
<i>Q7</i>						
<i>Q8</i>						
<i>Q9</i>						
<i>Q10</i>						

Code of Conduct:

*I __M.Mahathi(192324098)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

M.Mahathi

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS:

Total Marks: 10x2 =20

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

Q4. What is covariance analysis? List the two covariance formulas [L2, L3]

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set $x = \{4,5,6,7,9\}$, $y = \{8,3,7,5,6\}$

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, L5]

11/08/26

Class Test

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- 1) Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing.

Data Mining:

Data Mining is process of extracting useful, valid, and previously unknown patterns or knowledge from large datasets.

Any Two Data Mining Tasks: Classification, Clustering

Knowledge Discovery Databases Process.

1. Data Cleaning
2. Data Integration
3. Data Selection
4. Data Transformation
5. Data Mining
6. Pattern Evaluation
7. Knowledge Presentation.

Role of Data Preprocessing:-

Improves data quality by removing errors, handling missing values, and making data suitable for mining, leading to accurate results.

- 2) What is Data Cleaning? Mention data quality issues. Explain methods to handle missing values and noisy data.

Data Cleaning:

Data cleaning is process of detecting and correcting inaccurate or incomplete data.

Two Data Quality Issues : Missing Values , Noisy data

Handling Missing Values :

- Replace with mean / median / mode.
- Remove records with many missing values.

Handling Noisy Data:

- Binning
- Smoothing or clustering

3) Define Data Integration . Explain How redundancy and inconsistency are handled.

Data Integration :

Combining data from multiple sources into a single, unified dataset.

Challenges : Data redundancy , Data inconsistency

Handling Methods :-

- Redundancy : Remove duplicate records.
- Inconsistency : Standardize formats and resolve conflicting values.

4) What is Covariance Analysis ? List two covariance formulae.

Covariance Analysis :-

Covariance Measures how two variables change together.

Formulas:-

1. Population Covariance:

$$\text{Cov}(X, Y) = \frac{\sum (X - \mu_x)(Y - \mu_y)}{N}$$

2. Sample Covariance:

$$\text{Cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n - 1}$$

Interpretation:

- Positive $\rightarrow$ Move together
- Negative $\rightarrow$ Move oppositely
- Zero $\rightarrow$ No relationship.

5) Suppose two stocks A and B have values: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Will their prices rise or fall together or independently.

$$\text{Mean of } X \Rightarrow \frac{(2+3+5+4+6)}{5} = 4$$

$$\text{Mean of } Y \Rightarrow \frac{(5+8+10+11+14)}{5} = 9.6$$

X	Y	$X - 4$	$Y - 9.6$	product
2	5	-2	-4.6	9.2
3	8	-1	-1.6	1.6
5	10	1	0.4	0.4
4	11	0	1.4	0
6	14	2	4.4	8.8

Sum = 20

$$\text{population covariance} = \frac{20}{5} = 4$$

$$\text{Cov} = -10 \times 4 + 3 \times 4 - 3 \times 4 - 10 \times 4 + 10 \times 4 = -18$$

$$\text{Population Covariance} = \frac{\text{Cov}}{n} = \frac{-18}{9} = -2$$

Interpretation - It shows negative relationship exists between X and Y.

2) Importance the major steps in Data preprocessing are given as preprocessing pipeline.

Major steps

1) Data Cleaning - Data Integration, Data Transformation
2) Data Reduction - Data Compression

Reasons

- 1) Handling missing values
- 2) Remove duplicates and inconsistencies
- 3) Select and reject outliers
- 4) Normalise / transform data
- 5) Reduce dimensionality, data

Transformation

The process improves data quality before analysis.

1) Discrete data converting using Equal-Frequency (Equal-Width) Binning

Data (sorted) :

10, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30,
30, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Number of values = 27

Bin depth = 3

$$\text{count of Bin} = \frac{\text{Occurrence}}{\text{Bin depth}} = \frac{27}{3} = 9$$

Bin	Values	Mean
B1	13, 15, 16	$(13+15+16)/3 = 14.67$
B2	16, 19, 20	$(16+19+20)/3 = 18.33$
B3	20, 21, 22	$(20+21+22)/3 = 21.00$
B4	22, 25, 25	$(22+25+25)/3 = 24.00$
B5	25, 25, 30	$(25+25+30)/3 = 26.67$
B6	33, 33, 35	$(33+33+35)/3 = 33.67$
B7	35, 35, 35	$(35+35+35)/3 = 35.00$
B8	36, 40, 45	$(36+40+45)/3 = 40.33$
B9	46, 52, 70	$(46+52+70)/3 = 56.00$

Replace the mean for all 3 values in the Bin

Bin	original values	Bin Mean	Smoothed Values
B1	13, 15, 16	14.67	14.67, 14.67, 14.67
B2	16, 19, 20	18.33	18.33, 18.33, 18.33
B3	20, 21, 22	21.00	21.00, 21.00, 21.00
B4	22, 25, 25	24.00	24.00, 24.00, 24.00
B5	25, 25, 30	26.67	26.67, 26.67, 26.67
B6	33, 33, 35	33.67	33.67, 33.67, 33.67
B7	35, 35, 35	35.00	35.00, 35.00, 35.00
B8	36, 40, 45	40.33	40.33, 40.33, 40.33
B9	46, 52, 70	56.00	56.00, 56.00, 56.00

10) Explain Data Discretization and compare Equal-Width and Equal-Frequency Binning.

Data Discretization:

It converts continuous data into discrete intervals (bins).

Comparison

Equal-Width Binning	Equal - Frequency Binning
Equal interval size Simple to implement May create uneven bins <u>Ex</u> : - 0-10, 10-20, 20-30	Equal number of values Handles skewed data better Better data distribution <u>Ex</u> : - Each bin contains the same number of records.